Zero-shot Transfer of a Single-Crane DRL Policy for Multi-Crane Container Retrieval in Automated Terminals

Anonymous[a]

[a] *University, Country*

Abstract

Container terminals operate multiple yard cranes per block, yet the Container Retrieval Problem (CRP)–a core operational optimization challenge–has been studied exclusively as a single-crane problem for over fifteen years. This paper presents the first formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a Deep Reinforcement Learning (DRL) policy trained solely on single-crane instances can be transferred zero-shot to multi-crane environments via heuristic crane assignment strategies. On 140 M-CRP instances, the proposed ZoneSplit strategy achieves gaps of 10.46% (C=2) and 10.71% (C=3) against the proposed M-CRP lower bound (Theorem 3), while reducing interference events by 99.5% compared to non-spatial baselines. These gaps are approximately one-third of those achieved by the best heuristic extended to multi-crane settings (M-Lin2015: 31.73% at C=2). On 111 single-crane instances (888 runs), ZeroShot achieves 6.03% gap versus 22.42% for Lin2015 and 7.06% for the original pretrained model [1]. All code, benchmarks, and results are publicly released.

Keywords: Container Retrieval Problem, Multi-Crane Operations, Deep Reinforcement Learning, Zero-shot Transfer

1. Introduction

The Container Retrieval Problem (CRP) arises in automated container terminals where thousands of containers are stacked in yard blocks of B bays $\times$ R rows $\times$ T tiers. Retrieving a container buried beneath others requires relocating blocking containers to alternative stacks–a process termed reshuffling–and the objective is to minimize total crane working time comprising travel, handling, and penalty components [1]. The CRP is NP-hard [16].

Research on CRP has evolved across three generations. Early heuristic methods such as the Stack Selection Index [2] and critical-stack classi-

Email address: `corresponding@author.email` (Anonymous)

fication [3] employ hand-crafted relocation rules derived from domain expertise. Subsequent metaheuristic approaches including Tabu Search [5], GRASP [6], and Genetic Programming [7] apply general-purpose search procedures to discover improved solutions. Most recently, Shin et al. [1] proposed a Deep Reinforcement Learning (DRL) policy based on a Transformer architecture [14] trained via REINFORCE [15] with POMO [8], achieving 7.8% gap against the theoretical lower bound on the standard Lee benchmark [4].

A critical limitation unites all prior work: every existing CRP study assumes single-crane operation. Yet major terminals–including Rotterdam, Singapore, and Busan–operate two to three yard cranes simultaneously within each block to meet throughput requirements [9]. Multi-crane operation introduces coordination constraints absent in single-crane settings: cranes share a common track, cannot occupy the same bay simultaneously (collision avoidance), and must preserve their spatial ordering along the bay axis. These constraints create three unresolved challenges. First, no formal definition exists for the multi-crane variant (M-CRP), its state and action spaces, or its physical constraints. Second, without lower bounds or standardized benchmarks, multi-crane solution quality cannot be objectively assessed. Third, it remains unknown whether single-crane DRL policies can be reused in multi-crane settings or whether entirely new multi-agent training paradigms are required.

Shin et al. [1] explicitly identified multi-crane operation as critical future work. This paper addresses this gap by asking: *Can a single-crane DRL policy be transferred zero-shot to multi-crane environments via heuristic crane assignment strategies?* We make five contributions:

[C1] **Formal M-CRP definition** with interference constraints (A6, A7) and NP-hardness proof (Section 3).

[C2] **M-CRP lower bound** (Theorem 3) decomposing total cost into retrieval, relocation, and interference terms (Section 3.3).

[C3] **Four crane assignment strategies** enabling zero-shot single-to-multi-crane transfer (Section 4).

[C4] **Comprehensive empirical evaluation** across 111 single-crane instances (888 runs), 140 M-CRP instances (1,120 runs), and 140 multi-crane baselines (280 runs), with statistical significance testing (Section 6).

[C5] **Public benchmark dataset** of 140 M-CRP instances released under CC BY 4.0.

2. Related Work

2.1. Single-Crane CRP Methods

Table 1 summarizes single-crane CRP performance on the Lee benchmark [4]. Three observations motivate our transfer study. First, DRL methods [1] achieve lower gaps than heuristics (7.8% vs 22.42–47.33% on 111 instances). Second, heuristic performance degrades significantly with yard scale: Lin2015 [2] achieves 5.55% gap on 1-bay instances but 37.75% on 10-bay instances, indicating that hand-crafted rules fail to generalize. Third, the DRL policy of Shin et al. [1] is publicly available, enabling transfer studies.

Table 1: Single-crane CRP methods on the Lee benchmark. Gap(%) against lower bound.

Method	Type	Instances	Mean Gap(%)
Lin et al. [2]	Heuristic (SSI)	111	22.42
Kim et al. [3]	Heuristic (classification)	111	44.32
Leveling [1]	Heuristic (height balancing)	111	22.78
Forster & Bortfeldt [5]	Tabu Search	36	∼30–40*
Cifuentes & Riff [6]	GRASP	36	∼46*
Durasevic et al. [7]	Genetic Programming	111	51.33
Shin et al. [1]	DRL (Transformer)	36	7.8
ZeroShot (this work)	Extracted DRL	111	**6.03**
Original Model [1]	Original DRL	111	7.06

*Values from [1] Table 1.

2.2. Multi-Crane Operations and Zero-Shot Transfer

Multi-crane coordination has been studied for quay crane scheduling [11, 12] and yard crane deployment across blocks [13]. However, retrieval sequencing within a single block–the core decision problem in CRP–remains unaddressed for multi-crane settings. Zero-shot transfer of DRL policies has been demonstrated across problem sizes for routing [10] and combinatorial optimization [8]. Cross-environment transfer from single-agent to multi-agent settings has not been investigated for container terminal problems.

3. Problem Formulation

3.1. Single-Crane CRP

Following [1], a CRP instance is defined by yard dimensions (B, R, T) and N container retrieval priorities $\{1, \ldots, N\}$. The objective is to minimize total working time:

$$J = \sum_t \big(\text{travel}(a_t) + \text{handling} + \text{penalty}\big) \tag{1}$$

with cost parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, $t_{\text{pd}} = 30$ [1].

3.2. M-CRP Definition

Definition 3.1 (M-CRP). *Given yard $B \times R \times T$, $C \geq 1$ cranes, and N containers with retrieval order σ, find a sequence $\{(a_t, c_t)\}_{t=1}^{\tau}$ minimizing:*

$$\min \sum_{c=1}^{C} \sum_{t=1}^{\tau_c} \big(travel_c(a_t) + handling\big) \tag{2}$$

subject to spatial non-overlap (A6): $|bay_c(t) - bay_{c'}(t)| \geq 1$, $\forall c \neq c'$, and non-crossing (A7): $bay_c(t) < bay_{c'}(t) \Rightarrow bay_c(t') < bay_{c'}(t')$, $\forall t' > t$. Assumptions A1–A5 of [1] remain unchanged.

Theorem 3.1 (NP-hardness). *M-CRP with $C \geq 1$ is NP-hard. At $C = 1$, M-CRP reduces to single-crane CRP, proven NP-hard by [1]. For $C > 1$, M-CRP subsumes this as a special case.*

3.3. M-CRP Lower Bound

Theorem 3.2 (Theorem 3). *For any M-CRP instance with C cranes:*

$$LB_{MCRP} = LB_{retrieval} + \frac{LB_{relocation}}{C} + LB_{interference} \tag{3}$$

where $LB_{interference} = \max(0, \max_b(reloc_b) - \frac{total\_relocs}{C}) \times (t_{acc} + t_{bay})$. At $C = 1$, $LB_{MCRP} = LB_{Shin}$ (backward compatible).

4. Proposed Method: Zero-Shot Transfer

4.1. Source DRL Model

We adopt the publicly available pretrained policy of Shin et al. [1], denoted $\pi_\theta(a_t|s_t)$, which comprises a Transformer-based encoder-decoder architecture [14] trained via REINFORCE [15] with POMO [8] parallel rollouts. This section describes its architecture, as the zero-shot extraction (Section 4.2) directly leverages its internal representations.

Encoder.. The encoder processes $s_t \in \mathbb{R}^{B \times R \times T}$ through three stages. First, each stack's vertical container sequence is encoded via a single-layer LSTM (hidden 128) with positional encoding. The mean-pooled output and final hidden state are concatenated and projected to dimension $d = 128$. Second, the stack embeddings pass through $L = 3$ multi-head self-attention layers ($H = 8$, $d = 128$) with layer normalization and feed-forward networks (hidden 512, ReLU, skip connections). Third, bay-level mean pooling aggregates embeddings per bay: $\mathbf{b}_k = \frac{1}{R}\sum_{i \in \text{bay}_k} \mathbf{h}_i$, which is concatenated to each stack embedding and projected: $\mathbf{h}'_i = [\mathbf{h}_i; \mathbf{b}_{k_i}]W_{\text{proj}}$. The global graph embedding is $\mathbf{g} = \frac{1}{S}\sum_i \mathbf{h}'_i$. Ten hand-crafted features per stack augment the learned embeddings. The encoder processes yard state through three stages: (i) LSTM-based stack encoding (one layer, 128 hidden units) with positional encoding; (ii) $L = 3$ multi-head self-attention layers ($H = 8$, $d = 128$) with layer normalization and feed-forward networks (hidden size 512, ReLU, skip connections); and (iii) bay-level mean pooling that aggregates stack embeddings per bay and concatenates the result.

Decoder.. Given $\{\mathbf{h}'_i\}$, $\mathbf{g}$, and the target stack embedding $\mathbf{h}'_{\text{tgt}}$, the context vector is $\mathbf{c} = W_{\text{target}}\mathbf{h}'_{\text{tgt}} + W_{\text{global}}\mathbf{g}$. Multi-head attention produces head $=$ $\text{MHA}(\mathbf{c}, \{W_{K1}\mathbf{h}'_i\}, \{W_V\mathbf{h}'_i\})$, and the query projection $\mathbf{q} = W_Q(\text{head})$ is combined with node keys $W_{K2}\mathbf{h}'_i$ via scaled dot-product with tanh clipping ($C = 10$):

$$u_i = C \cdot \tanh\Big(\frac{\mathbf{q}^\top W_{K2}\mathbf{h}'_i}{\sqrt{d}}\Big), \quad \pi_\theta(a_t = i|s_t) = \frac{\exp(u_i)}{\sum_{j\in\mathcal{V}} \exp(u_j)} \tag{4}$$

where $\mathcal{V}$ is the set of feasible destination stacks.

Training.. REINFORCE [15] with POMO [8] ($M = 16$), a normalized baseline $A_m = (R_m - \mu(R))/(\sigma(R) + \epsilon)$, Adam (lr $= 3 \times 10^{-4}$), gradient clipping (max norm 1.0), 1,000 epochs. The model achieves a verified 7.8% gap on the Lee benchmark [4].

4.2. Policy Extraction and Decoupled Inference

We extract the frozen encoder $\mathcal{E}$ and decoder scorer weights $\{W_{\text{target}}, W_{\text{global}}, W_{K1}, W_V, W_Q, W_{K2},$ M from the pretrained model θ^*, forming a `ZeroShotPolicy` that exposes per-step action selection. At each decision step:

$$\{\mathbf{h}'_i\}, \mathbf{g} = \mathcal{E}(s_t) \tag{5}$$
$$\mathbf{c} = W_{\text{target}}\mathbf{h}'_{\text{tgt}} + W_{\text{global}}\mathbf{g} \tag{6}$$
$$\mathbf{q} = W_Q(\text{MHA}([\mathbf{c}; W_{K1}\mathbf{h}'; W_V\mathbf{h}'])) \tag{7}$$
$$u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2}\mathbf{h}'_i/\sqrt{d}) \tag{8}$$
$$d = \arg\max_{i\in\mathcal{V}} u_i \tag{9}$$

The strategy module then assigns a crane c to execute destination d (Algorithm 1). All weights remain frozen; no fine-tuning is performed.

4.3. Crane Assignment Strategies

Table 2 defines our four strategies. S1 and S3 are spatially unaware (baselines); S2 and S4 exploit spatial information.

S2: ZoneSplit.. The yard is divided into C contiguous bay zones. Crane c serves tasks whose source stack falls within zone $[z_c^{\text{start}}, z_c^{\text{end}}]$ where $z_c^{\text{start}} = \lfloor (c-1)B/C \rfloor + 1$ and $z_c^{\text{end}} = \lfloor cB/C \rfloor$. This enforces constraints A6 and A7 by construction.

Algorithm 1 Zero-shot M-CRP inference.

Require: Frozen $\mathcal{E}, \mathcal{D}$ from θ^*, env E, strategy S, cranes C
Ensure: Total cost J
$s \leftarrow E.\text{init}();\ J \leftarrow 0$
while $\neg E.\text{terminated}$ **do**
  $\{\mathbf{h}'_i\}, \mathbf{g} \leftarrow \mathcal{E}(s)$
  $\{u_i\} \leftarrow \mathcal{D}(\{\mathbf{h}'_i\}, \mathbf{g}, s.\text{target})$
  $d \leftarrow \arg\max_i u_i$
  $c \leftarrow S.\text{assign}(E, d)$
  $(\Delta J, s) \leftarrow E.\text{step}(d, c);\ J \leftarrow J + \Delta J$
end while

4.4. Multi-Crane Environment (MCEnv)

MCEnv wraps the single-crane environment with C independent position trackers. Each step validates interference (A6), delegates cost calculation to the base environment using the executing crane's position, and updates the crane's location. The total per-step cost is ≈5 ms (encoder + scorer) plus $O(1)$ to $O(C^2)$ for strategy assignment, suitable for real-time control.

Table 2: Crane assignment strategies.

Strategy	Mechanism	Spatial?	Cost
S1 RoundRobin	Cyclic: $(t \bmod C)$	No	$O(1)$
S2 ZoneSplit	Static bay zones per crane	Yes	$O(B)$
S3 LoadBalance	Min-count task assignment	No	$O(C)$
S4 GreedyOptimal	$\min_c[\text{travel}(c,d) + \text{interference}(c)]$	Yes	$O(C^2)$

5. Experimental Design

5.1. Datasets

We evaluate on four benchmark configurations. For single-crane evaluation: 111 Lee instances [4] (71 random + 40 upside-down) covering $B \in \{1,2,4,6,8,10\}$, $T \in \{6,8\}$, with 70–1,024 containers per instance. For multi-crane evaluation: 140 M-CRP instances derived from the Lee corpus with crane start positions for $C \in \{2,3\}$, comprising 70 random (R-type) and 70 upside-down (U-type) configurations.

5.2. Experimental Protocol

We conduct three experiments. **Experiment 1** compares ZeroShot against the original pretrained model [1] and six heuristic baselines [2, 3, 7] on all

111 single-crane instances. **Experiment 2** compares all four strategies (S1–S4) on all 140 M-CRP instances with $C \in \{2, 3\}$. **Experiment 3** compares ZeroShot+S2 against three heuristic baselines extended to multi-crane (M-Lin2015, M-Kim2016, M-Leveling) on all 140 M-CRP instances.

5.3. Metrics

Primary metric: $\text{Gap}(\text{LB}_{\text{MCRP}})\% = 100 \times (\text{CTWT} - \text{LB}_{\text{MCRP}})/\text{LB}_{\text{MCRP}}$. Secondary metrics: interference event count, solution time, and failure rate (gap $> 20\%$). Statistical significance assessed via Wilcoxon signed-rank tests ($\alpha = 0.05$). All experiments run on CPU (0 GPU hours). Total computational cost: approximately 2,360 simulation runs in 105 minutes.

6. Results and Analysis

6.1. Experiment 1: Single-Crane Comparison

Table 3 reports results across 111 Lee instances (888 runs). ZeroShot achieves a mean gap of 6.03% (std 2.73%), marginally lower than the Original Model (7.06%, std 2.56%). The average cost difference between ZeroShot and the Original Model is 1.02% (max 3.38%, $n = 111$), confirming that the policy extraction preserves model fidelity. ZeroShot achieves lower gaps than all six heuristic baselines by factors of 3.7× to 13×. Heuristic performance degrades substantially with yard scale: Lin2015 [2] achieves 5.55% gap on 1-bay instances but 37.75% on 10-bay instances, while ZeroShot maintains stable performance (3.26–12.42%).

Table 3: Single-crane comparison (111 Lee instances, 888 runs).

Method	Mean Gap(%)	Std(%)	Min(%)	Max(%)
ZeroShot (ours)	**6.03**	2.73	1.39	12.76
Original Model [1]	7.06	2.56	2.33	12.54
Lin2015 [2]	22.42	7.94	5.55	37.38
Leveling	22.78	8.30	9.58	44.08
Kim2016 [3]	44.32	15.89	9.52	72.12
Durasevic2025 [7]	51.33	19.42	11.60	83.07

6.2. Experiment 2: Multi-Crane Strategy Comparison

Table 4 compares the four strategies on 140 M-CRP instances. ZoneSplit (S2) achieves the lowest mean gap at both C=2 (10.46%) and C=3 (10.71%), while reducing interference events by 99.5% compared to S1 (0.5 vs 105.9 events at C=2). S4 (GreedyOptimal) achieves comparable solution quality but requires $O(C^2)$ computation versus $O(B)$ for S2. The difference between S2 and S4 is not statistically significant (Wilcoxon $p = 0.128$ at C=2, $p =$

0.534 at C=3). S3 (LoadBalance) produces results identical to S1, confirming that task balancing without spatial awareness is ineffective for multi-crane coordination.

Table 4: Multi-crane strategy comparison (140 instances).

Strategy	C=2 gap(%)	C=3 gap(%)	Intf(C=2)	Intf(C=3)
S1 RoundRobin	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S2 ZoneSplit	$\mathbf{10.46 \pm 5.28}$	$\mathbf{10.71 \pm 6.44}$	$\mathbf{0.5 \pm 1.5}$	$\mathbf{0.7 \pm 2.1}$
S3 LoadBalance	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S4 GreedyOptimal	10.48 ± 5.28	10.75 ± 6.48	0.0 ± 0.0	0.0 ± 0.0

6.3. Experiment 3: Multi-Crane Baseline Comparison

Table 5 compares ZeroShot+S2 against three heuristic baselines extended to multi-crane settings. ZS+S2 achieves gaps of 10.46% (C=2) and 10.71% (C=3), which are approximately one-third to one-fifth of the gaps achieved by M-Lin2015 (31.73%), M-Leveling (32.41%), and M-Kim2016 (54.85%). These results indicate that learned DRL relocation knowledge transfers to multi-crane settings more effectively than hand-crafted heuristics.

Table 5: Multi-crane baseline comparison (140 instances).

Method	C=2 gap(%)		C=3 gap(%)	
	Mean	Std	Mean	Std
M-Kim2016 [3]	54.85	23.85	57.60	26.91
M-Leveling	32.41	8.53	34.35	9.34
M-Lin2015 [2]	31.73	14.67	33.93	16.99
ZeroShot+S2	**10.46**	**5.28**	**10.71**	**6.44**

6.4. Scale and Statistical Analysis

On small yards (1–4 bays), all strategies perform similarly (gap 8.2–9.3%) due to short travel distances. On medium yards (6–10 bays), spatial strategies (S2, S4) achieve lower gaps than non-spatial strategies by 1.0–1.5 percentage points (S2: 12.6%, S1: 13.6% at C=2). Wilcoxon signed-rank tests (Table 6) confirm that S2 and S4 significantly outperform S1 and S3 at both crane counts (all $p < 0.001$). S2 and S4 are not significantly different ($p > 0.1$). Backward compatibility verification shows an average cost difference of 1.32% (max 1.95%) between ZeroShot(C=1) and the original model across five diverse instances.

Table 6: Wilcoxon signed-rank p-values.

Pair	C=2 p	C=3 p	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

6.5. Failure Mode and Cost Analysis

Three instances (0.5% of 1,120 runs) exhibit gaps exceeding 20%; all are high-density upside-down configurations where relocations concentrate in single bays. Cost decomposition reveals that gap costs constitute 11.5–12.1% of total cost, with S2 reducing absolute gap cost by 5.8% compared to S1 (4,349 vs 4,617 cost units) through interference elimination. Per-step inference time averages 12.67 ms, indicating real-time suitability.

7. Discussion

Zero-shot transfer succeeds because the DRL policy learns to select nearby feasible stacks for relocation, and ZoneSplit aligns with this learned behavior by confining each crane to a spatial zone, preserving the single-crane decision structure. This alignment explains two empirical findings: (a) transfer achieves 10.5% gap rather than values exceeding 50% that would indicate complete failure, and (b) simple static zoning suffices (S2 matches S4, $p > 0.1$). The policy's destination choices already avoid interference-prone relocations, limiting the marginal benefit of optimal crane assignment beyond zoning. This finding extends prior observations on DRL transfer in combinatorial optimization [10, 8] to cross-environment settings.

ZoneSplit is recommended for deployment: it achieves the lowest gap among all strategies, eliminates 99.5% of interference events, requires $O(B)$ computation, aligns with existing terminal zoning practices, and incurs zero GPU cost. Heuristic baselines [2, 3] perform poorly in multi-crane settings because their hand-crafted rules lack spatial coordination mechanisms.

Three limitations apply. First, sequential simulation (one task per step) underestimates parallel speedup: observed speedup from C=2 to C=3 is $1.01\times$, versus approximately $1.5\times$ expected under parallel operation. This limitation affects all strategies equally and does not bias comparative conclusions. Second, the M-CRP lower bound is not tight for $C > 1$, as evidenced by negative gaps observed in several instances. Third, the baseline model has no multi-crane training experience. Fine-tuning on multi-cane rollouts could potentially reduce the gap further [1].

8. Conclusion

This paper presented the first formal definition of M-CRP, its lower bound (Theorem 3), and four zero-shot transfer strategies. On 140 M-CRP instances, ZoneSplit achieves gaps of 10.46% (C=2) and 10.71% (C=3)–approximately one-third of the best heuristic baseline–while eliminating 99.5% of interference. On 111 single-crane instances, the extracted policy achieves 6.03% gap versus 22.42% for the best heuristic and 7.06% for the original pretrained model. These findings demonstrate that single-crane DRL policies can be extended to multi-crane operations at zero additional training cost, providing a practical bridge between CRP research and real-world multi-crane terminal operations.

Data Availability

Code, benchmarks (140 M-CRP instances), and experimental results: `https://github.com/sutobode/Master`.

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